

Elements of Probability Theory

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Probability in Dimension 1

Random experiment

An experiment which repeated under same fixed conditions produce different results is called random experiment.

Examples:

1. Rolling a cube-shaped fare die



2. Choosing an integer even number randomly



3. Measuring temperature



Probability space

The mathematical framework in which a random experiment is described consists of a triplet $(\Omega, \Sigma, \mathcal{P})$, called probability space.

The elements of a probability space are:

- Ω : sample space
(set with all possible events)
- Σ : σ -algebra on Ω
(set with relevant events only)
- $\mathcal{P}$: probability measure
(measure of expectation of an event occurrence)

Sample space

A non-empty set which contains all possible events for a certain random experiment is called sample space, being represented by Ω .

Examples:

1. Rolling a cube-shaped fare die (finite Ω)

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

2. Choosing an integer even number randomly (denumerable Ω)

$$\Omega = \{\dots, -8, -6, -4, -2, 0, 2, 4, 6, 8, \dots\}$$

3. Measuring temperature in Kelvin (non-denumerable Ω)

$$\Omega = [a, b] \subset [0, +\infty)$$

σ -algebra of events

In general, not all of the events in Ω are of interest.

Intuitively, a σ -algebra on Ω is the set of relevant outcomes for a random experiment. Formally, Σ is a σ -algebra on Ω if

- $\phi \in \Sigma$
(contains the empty set)
- $\mathcal{A}^c \in \Sigma$ for any $\mathcal{A} \in \Sigma$
(closed under complementation)
- $\bigcup_{i=1}^{\infty} \mathcal{A}_i \in \Sigma$ for any $\mathcal{A}_i \in \Sigma$
(closed under denumerable unions)

Probability measure

A probability measure is a function $\mathcal{P} : \Sigma \rightarrow [0, 1] \subset \mathbb{R}$ such that

- $\mathcal{P} \{ \mathcal{A} \} \geq 0$ for any $\mathcal{A} \in \Sigma$
(probability is nonnegative)
- $\mathcal{P} \{ \Omega \} = 1$
(entire space has probability one)
- $\mathcal{P} \left\{ \bigcup_{i=1}^{\infty} \mathcal{A}_i \right\} = \sum_{i=1}^{\infty} \mathcal{P} \{ \mathcal{A}_i \}$ for any $\mathcal{A}_i$ mutually disjoint
(σ -additivity)

Remark:

$\mathcal{P} \{ \phi \} = 0$ (empty set has probability zero)

An example in discrete probability



A fair coin is thrown twice.

The number of faces is of interest.



A. C. Morgado, J. B. Pitombeira, P. C. P. Carvalho, P. J. Fernandez, **Análise Combinatória e Probabilidade**, SBM, 2016

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Probability space 1:

$$\Omega_1 = \{(H,H), (H,T), (T,H), (T,T)\}$$



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$$\mathcal{P}_1 \{(H,H)\} = 1/4, \quad \mathcal{P}_1 \{(H,T)\} = 1/4,$$

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Probability space 2:

$$\Omega_2 = \{0, 1, 2\}$$



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Probability space 2:

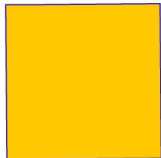
$$\Omega_2 = \{0, 1, 2\}$$

$$\mathcal{P}_2 \{0\} = 1/4, \quad \mathcal{P}_2 \{1\} = 1/2, \quad \mathcal{P}_2 \{2\} = 1/4$$



A. C. Morgado, J. B. Pitombeira, P. C. P. Carvalho, P. J. Fernandez, **Análise Combinatória e Probabilidade**, SBM, 2016

An example in continuous probability



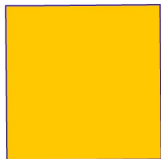
A point is randomly chosen in a square (side b).

Event A: point lie above main diagonal

Event B: point lie in main diagonal

Event C: point lie outside main diagonal

An example in continuous probability



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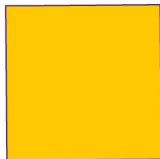
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$$\mathcal{P}\{A\} = \frac{\text{area of upper triangle}}{\text{area of square}} = \frac{0.5 b^2}{b^2} = \frac{1}{2}$$

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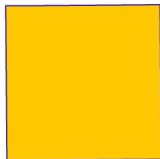
Event B: point lie in main diagonal

Event C: point lie outside main diagonal

$$\mathcal{P}\{A\} = \frac{\text{area of upper triangle}}{\text{area of square}} = \frac{0.5 b^2}{b^2} = \frac{1}{2}$$

$$\mathcal{P}\{B\} = \frac{\text{area of main diagonal}}{\text{area of square}} = \frac{0}{b^2} = 0$$

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$$\mathcal{P}\{C\} = \frac{\text{area outside main diagonal}}{\text{area of square}} = \frac{b^2 - 0}{b^2} = 1$$

Remarks on probability

Probability zero

- An impossible event has probability zero
(e.g. roll a six faces dice, numbered from 1 to 6, and get 7)
- Not every event with probability zero is impossible
(e.g. randomly pick a point on the main diagonal of a square)

Probability one

- An event which occurrence is certain has probability one
(e.g. throw a coin and obtain head or tail)
- Not every event with probability one occurs
(e.g. randomly pick a point outside square's main diagonal)

Conditional probability

Consider a pair of random events A and B such that $\mathcal{P}\{B\} > 0$.

The conditional probability of A , given the occurrence of B , denoted as $\mathcal{P}\{A|B\}$, is defined as

$$\mathcal{P}\{A | B\} = \frac{\mathcal{P}\{A \cap B\}}{\mathcal{P}\{B\}}.$$

It follows that

$$\mathcal{P}\{A \cap B\} = \mathcal{P}\{A | B\} \times \mathcal{P}\{B\}.$$

An example in conditional probability



Somebody rolls a pair of six-sided dice.

A = value rolled on die 1

B = value rolled on die 2

What is the probability that $A = 2$ given that $A + B \leq 5$?



Conditional probability — Wikipedia, The Free Encyclopedia, 2017.

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$$\mathcal{P}\{A\} = 6/36$$



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$A | A+B \leq 5$

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$$\mathcal{P}\{A|A+B \leq 5\} = 3/10$$



Conditional probability — Wikipedia, The Free Encyclopedia, 2017.

Independence of events

If the occurrence of an event B does not affect the occurrence of an event A one has

$$\mathcal{P}\{A \mid B\} = \mathcal{P}\{A\}.$$

In this way, once $\mathcal{P}\{A \cap B\} = \mathcal{P}\{A \mid B\} \times \mathcal{P}\{B\}$ it is true that

$$\mathcal{P}\{A \cap B\} = \mathcal{P}\{A\} \times \mathcal{P}\{B\}.$$

Events A and B in which the latter holds are said to be independent.

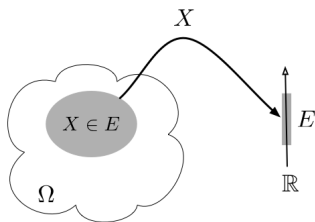
Remark:

This notion generalizes itself naturally to n events.

Random variable

A mapping $X : \Omega \rightarrow \mathbb{R}$ is called a random variable (RV) if the preimage of every real number under X is a relevant event, i.e.,

$$X^{-1}(x) = \{\omega \in \Omega : X(\omega) \leq x\} \in \Sigma, \quad \text{for every } x \in \mathbb{R}.$$



A collection of events in Ω is mapped to an interval E on the real line under such mapping.

©

RV are **numerical characteristics** of interesting events.

Remark:

A random variable is a function from Ω to $\mathbb{R}$, **not a real number**.

Examples of random variables

1. Rolling die experiment

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

- $$X(\omega) = \begin{cases} 1 & \text{if } \omega \text{ is even} \\ 0 & \text{if } \omega \text{ is odd} \end{cases}$$

(random variable)

$$\Sigma = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$$

2. Temperature (in Kelvin) measurement experiment

$$\Omega = [a, b] \subset [0, +\infty)$$

$$\Sigma = \mathcal{B}_{[a,b]} \text{ (Borel } \sigma\text{-algebra)}$$

- $$X(\omega) = -459.67 + 1.8 \omega$$

(random variable)



G. Grimmett and D. Welsh, **Probability: An Introduction**. Oxford University Press, 2 edition, 2014.

Probability distribution

The probability distribution of X , denoted by F_X , is defined as the probability of the elementary event $\{X \leq x\}$, i.e.,

$$F_X(x) = \mathcal{P}\{X \leq x\}.$$

F_X is also known as cumulative distribution function (CDF) and has the following properties:

- $0 \leq F_X(x) \leq 1$
- F_X is a monotonic, non decreasing, right continuous function
- $\mathcal{P}\{x_1 < X \leq x_2\} = F_X(x_2) - F_X(x_1) = \int_{x_1}^{x_2} dF_X(x)$
- $\int_{\mathbb{R}} dF_X(x) = 1$
- $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$

Probability density function

If the function F_X is differentiable, its derivative $p_X(x) = dF_X(x)/dx$ is called probability density function (PDF) of X , and one has

$$F_X(x) = \int_{-\infty}^x p_X(\xi) d\xi.$$

Note also that:

- $p_X(x) \geq 0$ for every $x \in \mathbb{R}$
- $\int_{\mathbb{R}} p_X(x) dx = 1$

Remark:

Intuitively, $p_X(x) dx$ can be thought of as the probability of X falling within the infinitesimal interval $[x, x + dx]$.

Types of random variables

1. Discrete random variable

Distribution is discrete.

Assumes a denumerable number of values.

Typically associated with counting processes.

2. Continuous random variable

Distribution is continuous.

Assumes a non-denumerable number of values.

Typically associated with measuring processes.

3. Mixed random variable

Distribution has points of discontinuity.

Assumes a non-denumerable number of values.

A “mixture” of the two previous types.

4. Singular random variable

Distribution is not differentiable at any point.

It has theoretical interest only.

Mathematical expectation operator

The mathematical expectation of a random variable X is defined as

$$\mathbb{E}\{X\} = \int_{\mathbb{R}} x \, dF_X(x).$$

The mathematical expectation is a **linear operator** since

$$\mathbb{E}\{\alpha_1 X_1 + \alpha_2 X_2\} = \alpha_1 \mathbb{E}\{X_1\} + \alpha_2 \mathbb{E}\{X_2\},$$

for any pairs of number α_1, α_2 and random variables X_1, X_2 .

Theorem (**law of the unconscious statistician**):

Given a measurable mapping $h : \mathbb{R} \rightarrow \mathbb{R}$ and a random variable X the expected value of $h(X)$ is given by

$$\mathbb{E}\{h(X)\} = \int_{\mathbb{R}} h(x) \, dF_X(x).$$

Mean value

The mean value of the random variable X is defined as

$$\begin{aligned}\mu_X &= \mathbb{E}\{X\} \\ &= \int_{\mathbb{R}} x dF_X(x) \\ &= \int_{\mathbb{R}} x p_X(x) dx.\end{aligned}$$

(measure of the central tendency)

Remark:

The mean value μ_X is the constant which best approximates the random variable X . The error of this approximation is the standard deviation σ_X .

Variance

The variance of the random variable X is defined as

$$\begin{aligned}\sigma_X^2 &= \mathbb{E} \left\{ (X - \mu_X)^2 \right\} \\ &= \int_{\mathbb{R}} (x - \mu_X)^2 dF_X(x) \\ &= \int_{\mathbb{R}} (x - \mu_X)^2 p_X(x) dx.\end{aligned}$$

(measure of dispersion about the mean)

Note that variance can also be written as

$$\sigma_X^2 = \mathbb{E} \left\{ X^2 \right\} - (\mathbb{E} \{X\})^2.$$

Remark:

σ_X^2 has the same unit as X^2 .

Standard deviation and variation coefficient

Other second-order statistics of X are the [standard deviation](#)

$$\sigma_X = \sqrt{\sigma_X^2},$$

and the [variation coefficient](#)

$$\delta_X = \sigma_X / \mu_X, \quad \mu_X \neq 0.$$

(both are measures of dispersion about the mean)

Remark:

σ_X has the same unit as X and δ_X is dimensionless.

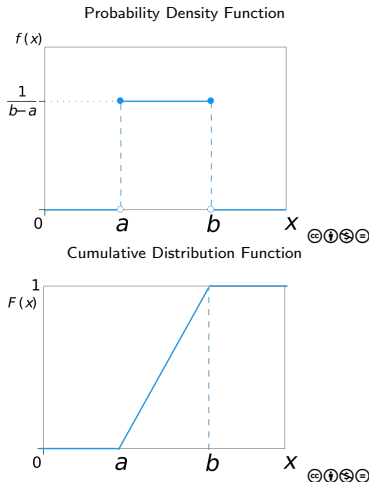
Probability Distributions

Uniform

- Notation: $\mathcal{U}(a, b)$
- Support: $[a, b]$
- Parameters:
 - $-\infty < a < b < +\infty$ — boundaries
- PDF:

$$p_X(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

- Statistics:
 - $\mu = \frac{1}{2}(a+b)$
 - $\sigma^2 = \frac{1}{12}(b-a)^2$



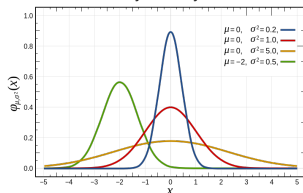
Uniform distribution — Wikipedia, The Free Encyclopedia, 2021.

Gaussian

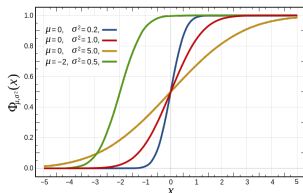
- Notation: $\mathcal{N}(\mu, \sigma^2)$
- Support: $(-\infty, +\infty)$
- Parameters:
 - μ — mean
 - σ^2 — variance
- PDF:

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

Probability Density Function



Cumulative Distribution Function



Normal distribution — Wikipedia, The Free Encyclopedia, 2021.

Characterization of a probability distribution

Given the mean μ and standard deviation σ of a random variable.

Is the distribution well-defined?

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$$p_X(x) = \frac{1}{\sqrt{6}\pi} \exp\left\{-\frac{(x-1)^2}{6}\right\} \quad \text{and} \quad p_X(x) = \frac{1}{6} \mathbb{1}_{[-2,4]}(x)$$

have $\mu = 1$ and $\sigma = \sqrt{3}$.

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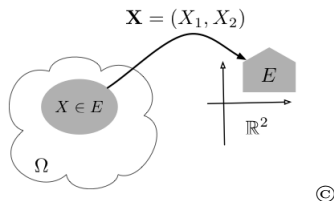
What type of information determines a distribution?

- cumulative distribution function
- probability density function (if exists)
- quantile function
- characteristic function
- moment-generating function (if exists)

Probability in Dimension n

Random vector

Let $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$. A random vector $\mathbf{X} = (X_1, \dots, X_n)$ is a collection of n random variables $X_i : \Omega \rightarrow \mathbb{R}$ that together may be considered a (measurable) mapping $\mathbf{X} : \Omega \rightarrow \mathbb{R}^n$.



A collection of event in Ω is mapped into a region on the Euclidean space under such mapping.

Joint probability distribution

The joint probability distribution of random vector $\mathbf{X} = (X_1, \dots, X_n)$, denoted by $F_{\mathbf{X}}$, is defined as

$$F_{\mathbf{X}}(x_1, \dots, x_n) = \mathcal{P} \{ \{X_1 \leq x_1\} \cap \dots \cap \{X_n \leq x_n\} \}.$$

Thus,

$$\mathcal{P} \{ \mathbf{a} < \mathbf{X} \leq \mathbf{b} \} = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} dF_{\mathbf{X}}(x_1, \dots, x_n),$$

in which $\{ \mathbf{a} < \mathbf{X} \leq \mathbf{b} \} = \{a_1 < X_1 \leq b_1\} \cap \dots \cap \{a_n < X_n \leq b_n\}$.

$F_{\mathbf{X}}$ is also known as joint cumulative distribution function.

Joint probability density function

If $p_{\mathbf{X}}(x_1, \dots, x_n) = \partial^n F_{\mathbf{X}} / \partial x_1 \cdots \partial x_n$ exists, for any $x_1, \dots, x_n$, then it is called joint probability density function of $\mathbf{X}$, and one has

$$F_{\mathbf{X}}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} p_{\mathbf{X}}(\xi_1, \dots, \xi_n) d\xi_1 \cdots d\xi_n.$$

Note also that:

- $p_{\mathbf{X}}(x_1, \dots, x_n) \geq 0$ for every $(x_1, \dots, x_n) \in \mathbb{R}^n$
- $\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p_{\mathbf{X}}(x_1, \dots, x_n) dx_1 \cdots dx_n = 1$

Marginal probability density function

The marginal probability density function of X_i is defined as

$$p_{X_i}(x_i) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1 \text{ times}} p_{\mathbf{X}}(x_1, \cdots, x_n) dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n,$$

for $i = 1, \cdots, n$.

Conditional distribution

Consider a pair of jointly distributed random variables X and Y . The conditional distribution of X , given the occurrence of the value y of Y , is defined as

$$F_{X|Y}(x | y) = \frac{F_{X,Y}(x, y)}{F_Y(y)}.$$

Thus

$$F_{X,Y}(x, y) = F_{X|Y}(x|y) \times F_Y(y),$$

and

$$p_{X,Y}(x, y) = p_{X|Y}(x|y) \times p_Y(y).$$

Remark:

This definition extends naturally to the n -dimensional case.

Independence of distributions

The random variables X and Y are said to be independent if the realization of X does not affect the probability distribution of Y , i.e.,

$$F_{X|Y}(x|y) = F_X(x).$$

Therefore, for independent random variable one has

$$F_{X \ Y}(x, y) = F_X(x) \times F_Y(y),$$

and

$$p_{X \ Y}(x, y) = p_X(x) \times p_Y(y).$$

Remark:

This definition extends naturally to the n -dimensional case.

Statistics of random vectors

- second-order random vector

$$\mathbb{E} \left\{ \|\mathbf{X}\|^2 \right\} = \int_{\mathbb{R}^n} \|\mathbf{x}\|^2 dF_{\mathbf{X}}(\mathbf{x}) < +\infty$$

- mean vector

$$\mathbf{m}_{\mathbf{X}} = \mathbb{E} \{ \mathbf{X} \} = \int_{\mathbb{R}^n} \mathbf{x} dF_{\mathbf{X}}(\mathbf{x}) \in \mathbb{R}^n$$

- correlation matrix

$$[R_{\mathbf{XY}}] = \mathbb{E} \left\{ \mathbf{XY}^T \right\} = \int_{\mathbb{R}^n} \mathbf{xx}^T dF_{\mathbf{XY}}(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{n \times n}$$

- covariance matrix

$$\begin{aligned} [K_{\mathbf{XY}}] &= \mathbb{E} \left\{ (\mathbf{X} - \mathbf{m}_{\mathbf{X}}) (\mathbf{Y} - \mathbf{m}_{\mathbf{Y}})^T \right\} \in \mathbb{R}^{n \times n} \\ &= [R_{\mathbf{XY}}] - \mathbf{m}_{\mathbf{X}} \mathbf{m}_{\mathbf{Y}}^T \end{aligned}$$

Remark:

Matrices $[R_{\mathbf{XY}}]$ and $[K_{\mathbf{XY}}]$ are symmetric positive semi-definite when $\mathbf{X} = \mathbf{Y}$.

Correlation of random variables

The random vectors $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{Y} = (Y_1, \dots, Y_n)$ are said to be uncorrelated if covariance matrix $[K_{\mathbf{XY}}]$ is null, i.e.,

$$[R_{\mathbf{XY}}] = \mathbf{m}_{\mathbf{X}}\mathbf{m}_{\mathbf{Y}}^T.$$

If two random vectors are independent, then they are uncorrelated.

independence $\implies$ uncorrelation

But uncorrelated random vectors are not independent in general.

uncorrelation $\nRightarrow$ independence

Remark:

Uncorrelated random vectors which the joint distribution is Gaussian are independent.

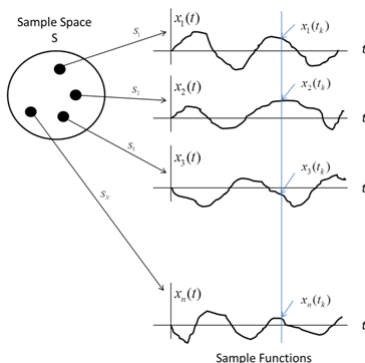
Notions of Random Processes

Random process

A real-valued random process (also called stochastic process) defined on probability space $(\Omega, \Sigma, \mathcal{P})$, indexed by $t \in \mathcal{T}$, is a mapping

$$(t, \omega) \in \mathcal{T} \times \Omega \rightarrow X(t, \omega) \in \mathbb{R},$$

such that, for fixed t , the output is a random variable $X(t, \cdot)$, while for fixed ω , $X(\cdot, \omega)$ is a function of t (sample function).



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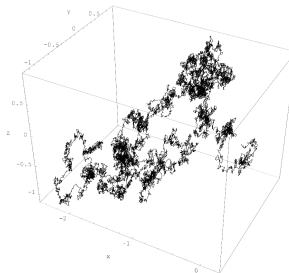
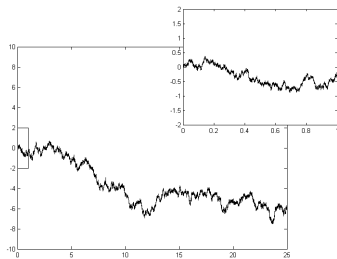
Interpretation and classification

Note that, by definition, a real-valued random process $X(t, \omega)$ is a collection of real-valued random variables indexed by a parameter, and can be thought of as a time-dependent random variable.

According to the nature of the set of indices $\mathcal{T}$, a stochastic process is classified as:

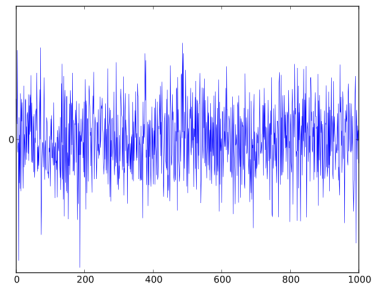
- discrete – if $\mathcal{T}$ is countable
- continuous – if $\mathcal{T}$ is uncountable

Wiener process (Brownian motion process)

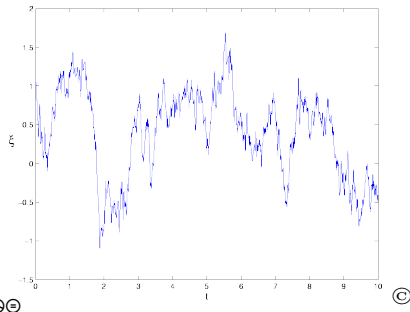


Wiener process — Wikipedia, The Free Encyclopedia, 2021.

White noise and colored noise



White noise



Colored noise



White noise — Wikipedia, The Free Encyclopedia, 2021.

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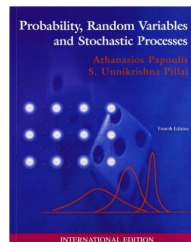
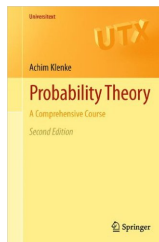
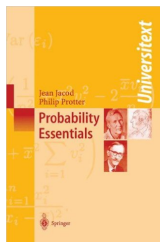
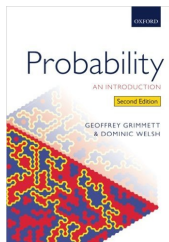
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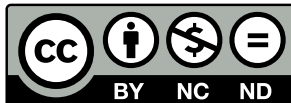


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